Spectral harmonicity as a candidate observable of proximity to criticality in models and human EEG

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Abstract

Criticality — the idea that cortex operates near a phase transition between order and disorder — is a leading candidate organizing principle for brain dynamics, but its standard markers are demanding to measure and frequently disagree: avalanche branching estimators need population recordings, long-range temporal correlations need extended stationary epochs, and susceptibility-based measures need many channels. Here we ask whether a single spectral property — the harmonicity H of a power spectrum, a phase-blind measure of how nearly the spectral peaks fall into integer-ratio relationships — behaves as a *candidate* observable of proximity to criticality. Across three model systems with independent ground truth, H peaked when the system was tuned to its critical point: a branching network at its critical branching ratio, a noise-driven echo-state reservoir at the edge of chaos, and a Wilson–Cowan excitatory/inhibitory network at the edge of synchronization. A surrogate battery showed this peak is a property of the power spectrum rather than of phase, and is not reproduced by slope-, power- and peakiness-matched null spectra — although it reflects the emergence of structured spectral peaks at criticality rather than exact integer-ratio tuning. We then tested the prediction in human EEG. A naive test reversed the model result; the prediction was recovered only when H was read off the model-matched, scale-free population observable, within sleep state and controlling for slow-wave power, where it tracked the wake-to-sleep traversal of criticality. The effect was modest and rested on this sleep traversal; propofol sedation and deep anaesthesia, which barely traverse the critical region according to our estimator, were boundary nulls. Finally, in a spiking network where avalanches and oscillations coexist, harmonicity read off the population signal tracked the emergent synchronization rather than the avalanche-critical point—the same observable-mixing that drives the raw-EEG reversal—so the criticality signal in H is exposed only on a scale-free observable, while $R = H \cdot \mathrm{PC}$ is its oscillation-gated synchronization index. H is thus a candidate, single-signal index of proximity to criticality whose in-vivo evidence remains preliminary and conditional, but whose model-side construct validity is strong and surrogate-controlled.

Introduction

A long-standing conjecture holds that the cerebral cortex operates near a critical point — a phase transition separating ordered, low-entropy dynamics from disordered, high-entropy dynamics — and that this proximity confers functional advantages such as maximal dynamic range, optimal information transmission, and large susceptibility to inputs (Bak et al., 1987; Beggs & Plenz, 2003; Chialvo, 2010; Shew & Plenz, 2013; Cocchi et al., 2017). On this view, criticality is not an incidental feature but a candidate organizing principle of cortical computation, with the brain poised at, or slightly below, the edge of an instability so that it can be both stable and maximally responsive (Priesemann, 2014). A closely related framing places the operating point at the edge of synchronization, where weakly coupled oscillators sit between incoherence and global phase-locking (di Santo et al., 2018; Fontenele, 2019). Whether read as an edge of a branching process or an edge of synchronization, the central claim is the same: dynamically rich behaviour emerges in a narrow region of parameter space, and the brain finds its way there.

The difficulty is that the observables used to certify criticality are demanding and do not always agree. The most direct estimator, the branching ratio of neuronal avalanches, requires population spiking or otherwise carefully subsampled recordings, and unbiased estimation of it is itself a methodological undertaking (Wilting & Priesemann, 2018; Spitzner, 2021; Priesemann, 2014). Long-range temporal correlations, quantified by detrended fluctuation analysis, require long, stationary epochs to yield a stable scaling exponent (Hardstone, 2012; Linkenkaer-Hansen et al., 2001). Susceptibility- and variance-based markers require many simultaneous channels to capture the divergence of fluctuations near the transition. Each marker carries its own assumptions, its own data requirements, and its own failure modes, and in the same recording they can point in different directions, so that whether a system is judged near-critical can depend as much on the chosen observable as on the system itself (O'Byrne & Jerbi, 2022; Cocchi et al., 2017). This proliferation of partially discordant markers motivates a search for an observable that is cheap to measure — ideally from a single channel, without strong stationarity assumptions and without a fitted model of the dynamics — yet still tracks proximity to the critical point.

We pursue such an observable in the structure of the power spectrum itself. Near a critical point, fluctuations become correlated across many temporal scales, and multi-scale structure of this kind can seed approximately integer-ratio relationships between spectral components — harmonic relationships — even in the absence of a single dominant oscillator. This suggests *harmonicity*, a measure of how nearly the prominent features of a spectrum fall into small-integer ratios, as a candidate read-out of criticality. Harmonicity has the properties we want: it is computed per frequency, it is phase-blind, and it can be estimated from one channel of data. We denote it H. Because harmonicity can be evaluated either over the full spectrum or over its oscillatory peaks, it admits an oscillation-gated companion, $R = H \cdot \mathrm{PC}$, in which H is multiplied by an index of peak prominence; R asks whether harmonic structure is carried specifically by genuine oscillations rather than by broadband structure.

Harmonicity as a spectral construct, and the distinction between genuine harmonic relationships and the spurious harmonics produced by waveform nonlinearity or cross-frequency coupling, are not new: they have been developed in work on spectral harmonicity and the template-based line indices used to separate the two (Dellavale et al., 2020; Bellemare-Pepin & Jerbi, 2025), in observations of harmonic peak organization in human EEG (Rodriguez-Larios et al., 2019), and, much earlier, in the harmonics-to-noise ratio of voice analysis (Boersma, 1993). Our contribution is not to introduce harmonicity but to operationalize it as a criticality observable and to test that interpretation against ground truth. The conceptual bridge runs through the edge-of-chaos literature, in which computational capacity — memory and nonlinear processing — is maximized in a narrow regime between order and chaos (Langton, 1990; Bertschinger, 2004; Jaeger & Haas, 2004); if the spectral signature of that regime is enhanced harmonicity, then H should peak there. Where oscillations and weak coupling are involved, the relevant theory is that of synchronization and mode-locking, in which integer-ratio (Arnold-tongue) relationships arise generically between coupled oscillators (Arnold, 1965; Pikovsky et al., 2001; Kuramoto, 1984; Glass & Mackey, 1988); our analyses recover this theory rather than discover it.

This paper makes three contributions. First, we show that harmonicity H indexes proximity to criticality across three model systems with independent ground truth — a branching network, a noise-driven reservoir, and an excitatory/inhibitory Wilson–Cowan network — and, on the appropriate observable, in human EEG; the near-ceiling model effects are evidence of construct validity rather than of effect magnitude. Second, we establish a division of labour between H and its oscillation-gated companion R: H is the general observable, peaking at criticality even in non-oscillatory systems, whereas R is an oscillation-gated synchronization index: it becomes placeable near a synchronization transition but indexes synchronization rather than avalanche criticality, and sits at the floor in non-oscillatory systems. Third, we resolve an apparent contradiction in vivo: a naive test of the prediction reverses, and the model relationship is recovered only when H is read off the model-matched, scale-free observable, within sleep state and controlling for slow-wave power. That recovery is modest and rests on the wake-to-sleep traversal of criticality; conditions that barely move the system across the critical region — propofol sedation and deep anaesthesia — are boundary nulls, and H and R do not outperform band power. We report these limits plainly, because they delineate exactly where a single-channel, model-free harmonicity read-out is, and is not, informative about how close cortex sits to criticality.

It helps to keep three axes distinct. *Avalanche/branching criticality* is the approach to a branching ratio $\hat{m} \to 1$ with scale-free avalanches; *synchronization* is the onset of collective oscillation at a Hopf-like transition; and the *observable* is the signal from which H is read. As we will show, H tracks avalanche criticality only when computed on a scale-free population observable; when a strong oscillation dominates

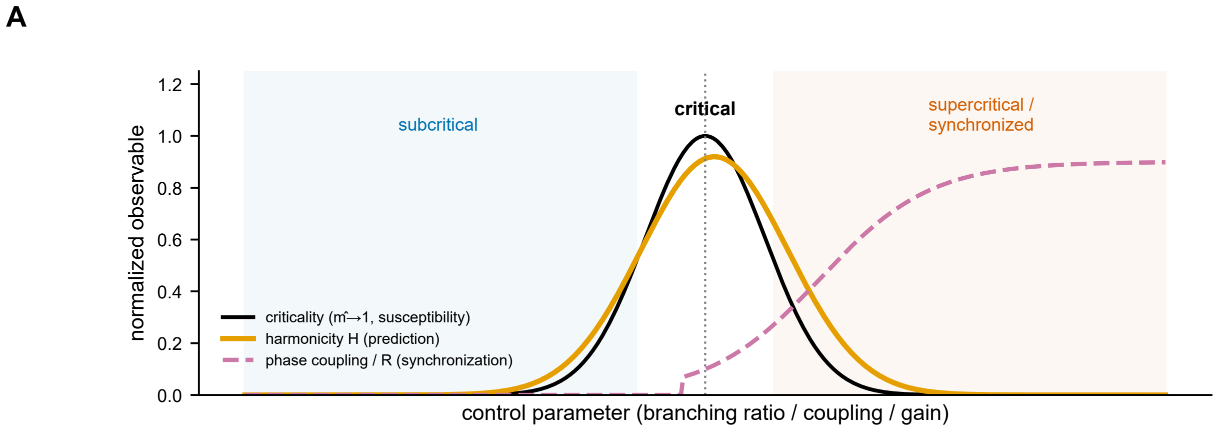

Figure 1. The hypothesis. As a system crosses its critical point, classical markers (branching ratio $\hat{m} \rightarrow 1$, susceptibility) peak; we predict that spectral harmonicity H peaks there too, whereas phase coupling and $R = H \cdot \mathrm{PC}$ rise with synchronization (supercritical / synchronized regime) and index the synchronization axis rather than avalanche criticality per se.

the signal, H instead follows the oscillatory harmonic series. R, by construction, indexes the synchronization axis and is not expected to track avalanche criticality. Conflating these axes is the source of the apparent in-vivo contradiction we resolve below.

Results

Harmonicity peaks at the critical point of a branching network

We first asked whether harmonicity behaves as a criticality observable in a system where the location of the critical point is known exactly. We used a probabilistic branching network, the canonical minimal model of neuronal avalanches (Beggs & Plenz, 2003; Bak et al., 1987), and swept its control parameter σ (the expected number of descendants per active unit) through its critical value at $\sigma = 1$. By construction, the network is subcritical for $\sigma < 1$, critical at $\sigma = 1$, and supercritical for $\sigma > 1$, which lets us compare any candidate observable against unambiguous ground truth.

The classical signatures behaved as expected and gave us a fixed reference for the critical point. Susceptibility (the variance of the population activity) peaked at $\sigma \approx 1$, the avalanche size distribution was best described by a power law there (the goodness-of-fit R^2 for the power-law regime peaked at $\sigma \approx 1$), and the branching ratio $\hat{m}$ recovered from the activity, estimated with the multistep regression method (Wilting & Priesemann, 2018; Spitzner, 2021), crossed unity at the same point. These three independent markers therefore agreed on where criticality lies.

We then read spectral harmonicity off the same activity. Harmonicity $H_{\max}$ traced a clear inverted-U across the sweep, rising from the subcritical regime, reaching its maximum at the critical point, and falling again into the supercritical regime. The peak sat at $\sigma = 1.00$ (95% CI $[0.97, 1.08]$), squarely on the critical point fixed by the classical markers. In other words, the activity carried the most harmonically organized spectral structure exactly where the network was poised at criticality, and lost that structure on either side.

To establish that this co-location is a property of the system rather than a coincidence of a single sweep, we tested it two ways. First, we compared the location of the H peak to the location of the susceptibility peak directly: the bootstrap difference between the two peak positions was -0.05 (95% CI $[-0.11, +0.00]$), an interval that contains zero, so the harmonicity peak and the susceptibility peak are statistically indistinguishable in location. Second, we asked whether harmonicity tracks proximity to criticality network-by-network rather than only on average. Across 20 independently realized networks, per-network $H_{\max}$ scaled with criticality proximity, quantified as $-|\hat{m} - 1|$, with a strong rank correlation (Spearman $\rho = +0.82$, 95% CI $[0.78, 0.86]$, $p = 1.9 \times 10^{-6}$); the relationship held in 100% of the 20 networks. Harmonicity is thus not

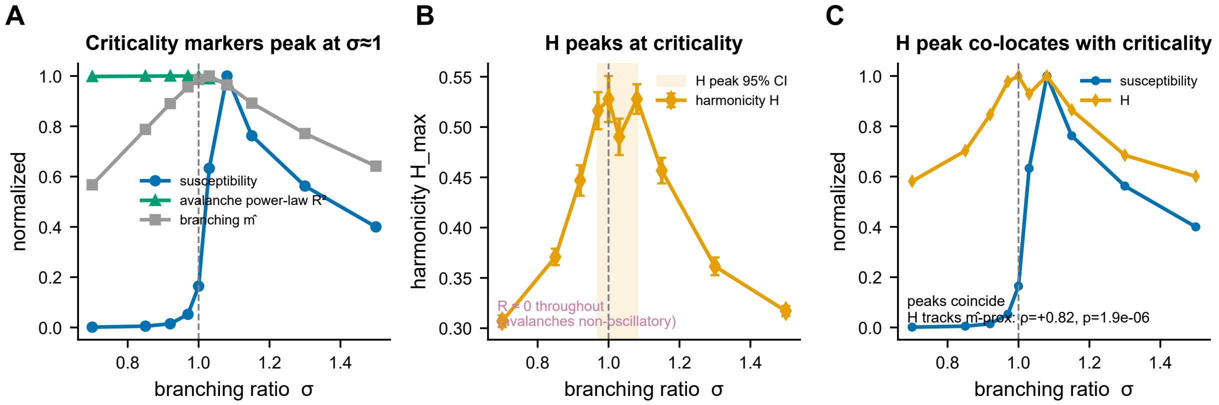

Figure 2. **Harmonicity peaks at criticality in a branching network.** **(A)** Classical criticality markers (susceptibility, avalanche power-law R^2, branching estimate $\hat{m}$) peak at $\sigma \approx 1$. **(B)** Harmonicity $H_{\max}$ is an inverted-U peaking at the critical point (shaded 95% CI of the peak location); phase-coupling R stays at the floor because bare avalanches are scale-free but non-oscillatory. **(C)** The H peak co-locates with the susceptibility peak, and H tracks criticality proximity across σ.

merely peaked at the group level but monotonically reports how close each individual network sits to its own critical point (Figure 2).

Finally, this experiment makes plain which observable carries the signal. The oscillation-gated companion $R = H \cdot \mathrm{PC}$ remained pinned at its $\sim 10^{-5}$ floor throughout the sweep. This is exactly what the construction predicts: bare avalanches are scale-free but not oscillatory, so the phase-coupling gate PC has nothing to lock onto and suppresses R everywhere, including at criticality. The criticality signature therefore lives in H, not in R. This is the central honest finding of the model series: H, not R, is the criticality observable, and R adds value only once genuine oscillatory structure is present.

Criticality generates harmonicity from structureless noise

Avalanche dynamics are scale-free but silent in the oscillatory channel, so a single model cannot tell us whether harmonicity merely indexes power-law statistics or whether a near-critical system actively builds harmonic structure. We therefore turned to a second, very different system: an echo-state reservoir, a high-dimensional recurrent network whose dynamics pass through an edge of chaos as recurrent gain increases (Jaeger & Haas, 2004; Langton, 1990; Bertschinger, 2004). Crucially, we drove the reservoir with structureless white noise. Any harmonic organization in its emergent activity therefore cannot have been inherited from the input; it must be generated internally.

We located the edge of chaos independently of harmonicity using the largest Lyapunov exponent, which changes sign as trajectories switch from contracting to expanding. The Lyapunov estimate placed the edge at a spectral radius $\rho_c = 1.57$ (95% CI $[1.44, 1.68]$), giving us a ground-truth marker for the dynamical transition against which to test the spectral observable.

We then measured how much harmonic structure the reservoir's emergent mode carried, expressed as a gain over the flat-noise input baseline so that zero gain means the output is no more harmonically organized than the white-noise drive. The reservoir generated harmonic structure well above this baseline: the gain became significantly positive at an onset of $\rho = 1.30$ and peaked at $+0.15$ (95% CI $[0.06, 0.25]$) near $\rho \approx 1.5$. The harmonicity gain therefore rose and peaked in the self-sustaining regime around and just past the edge of chaos located by the Lyapunov exponent, precisely where the recurrence becomes strong enough to maintain its own activity. A system fed only noise thus manufactures harmonic spectral structure when it is poised near its critical transition, and this generation is read off the single-channel observable H exactly as in the branching network (Figure 3).

We also checked whether this generative effect is just a relabeling of the reservoir's well-known computational optimum. It is not. Memory capacity, the standard benchmark of reservoir computation, peaked

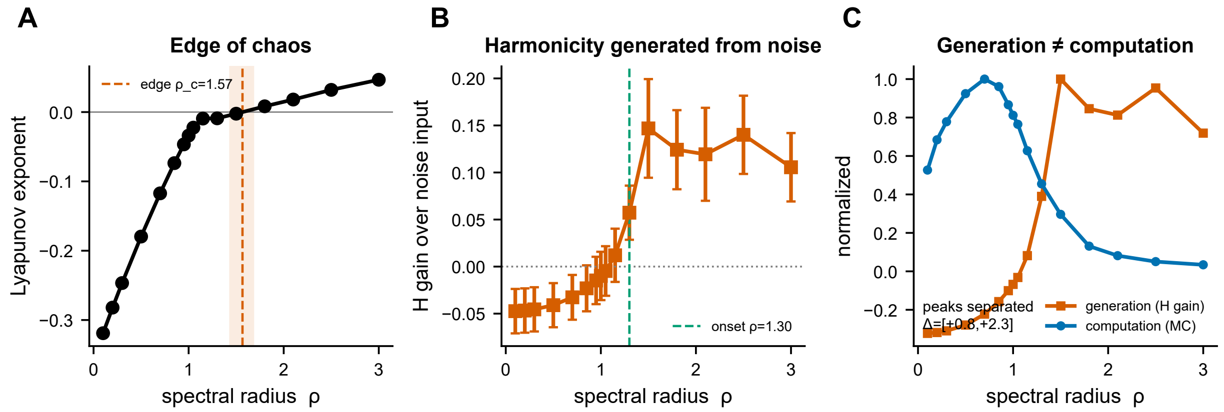

Figure 3. Criticality generates harmonicity from noise. (A) The Lyapunov exponent locates the edge of chaos ρ_c (zero-crossing, 95% CI). **(B)** Driven by white noise, the reservoir's emergent mode gains harmonic structure over the (flat) input near and beyond the edge (generation onset marked). **(C)** Generation (harmonicity gain) dissociates from computation (memory capacity): their peaks fall at different ρ.

far below the edge, at $\rho \approx 0.7$, deep in the stable regime. The harmonicity-gain peak was clearly separated from the memory-capacity peak, with a bootstrap peak difference of +1.27 (95% CI $[0.80, 2.30]$), an interval well clear of zero. Harmonic generation and computational performance therefore optimize at different operating points: harmonicity reports proximity to the dynamical edge, not the regime that maximizes task memory. Together with the branching network, this establishes harmonicity as a model-free observable that both peaks at criticality when scale-free structure is imposed and is actively generated near criticality when only noise is supplied.

At the synchronization onset, phase coupling becomes placeable

Does R become placeable when oscillations and a synchronization transition coexist? The three preceding models established H as a robust marker but left $R = H \cdot \mathrm{PC}$ at the floor, because none of them generated sustained oscillations for the phase-coupling term to register. To place R, we required a system in which a synchronization transition and a critical point are co-located. We therefore simulated a stochastic Wilson–Cowan excitatory–inhibitory network and swept the recurrent gain g across its Hopf bifurcation, the onset of collective oscillation. Such E/I networks are the canonical setting in which an edge of synchronization coincides with a critical transition between cortical states (di Santo et al., 2018; Fontenele, 2019), making them the natural home for an oscillation-gated marker.

Identifying the critical point demanded care, because the most obvious marker is misleading. The raw susceptibility, measured as the variance of the oscillation envelope, peaked sharply at the synchronization onset $g_c = 1.0$. That peak, however, is amplitude-confounded: the envelope grows in absolute scale as the network entrains, so its variance inflates for reasons that have nothing to do with critical fluctuations. When we instead computed the normalized susceptibility (variance divided by mean^2), which removes this scale dependence, the peak moved earlier, to $g \approx 0.60$ (95% interval $[0.50, 0.60]$). This earlier location is the one that agrees with the model-independent signature of criticality: the autocorrelation time, our measure of critical slowing down, peaked at the same gain. Critically, this critical point is fixed independently by the deterministic system itself: the leading eigenvalue of the noise-free Wilson–Cowan Jacobian crosses zero—the Hopf bifurcation—at $g = 0.66$, computed with no reference to harmonicity or resonance and coinciding with the normalized-susceptibility and critical-slowing peaks. The disagreement between markers was thus resolved in favor of the normalized measure, and it is at this normalized-susceptibility peak, not at the raw-amplitude peak, that the true critical region sits.

With the critical region fixed independently, we asked where the resonance observables sit. This is the one of the three models in which R is non-trivial: harmonicity H rises and phase coupling switches on at the

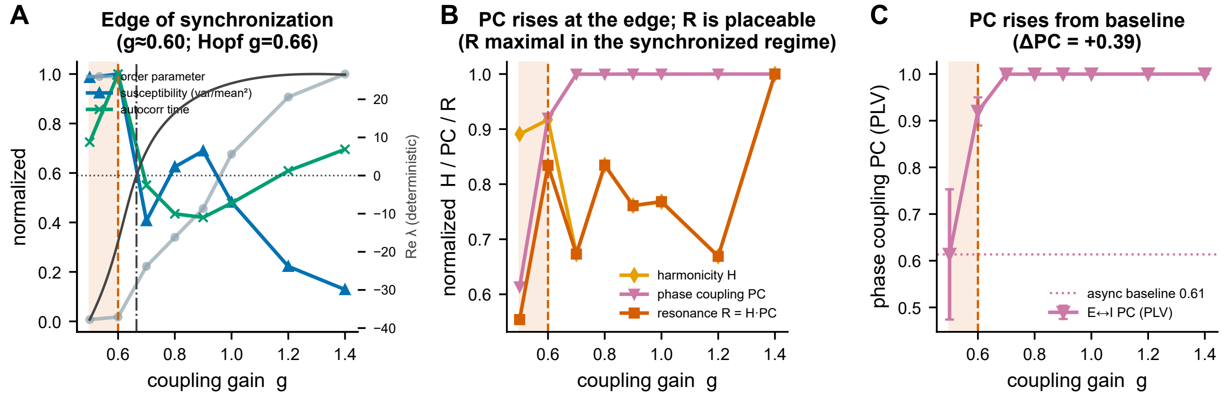

Figure 4. At the synchronization onset, phase coupling and resonance become placeable. (A) The edge of synchronization, located by the relative-fluctuation susceptibility (var/mean^2) and the autocorrelation time (critical slowing), which peak together near $g \approx 0.60$ (shaded), where the deterministic Jacobian's leading eigenvalue also crosses zero (Hopf bifurcation, $g = 0.66$); raw envelope-variance susceptibility peaks later, an amplitude confound. **(B)** The resonance factors against the edge: harmonicity H, phase coupling PC, and $R = H \cdot \mathrm{PC}$ (normalized); PC rises through the edge and R becomes non-trivial here—the only one of the three models where it does—though R is maximal in the synchronized regime above the edge, not exactly at it. **(C)** The absolute rise of E↔I phase coupling (PLV) above its non-zero asynchronous baseline (ΔPC = +0.39).

onset, so $R = H \cdot \mathrm{PC}$ becomes placeable rather than sitting at the floor. R does *not* sharply peak at the edge, however—it rises through the onset and stays elevated across the synchronized regime, with its maximum well into synchrony—foreshadowing the spiking-network result below that R indexes synchronization rather than the avalanche-critical point per se. The phase-coupling term that gates R behaved as a graded quantity rather than a switch: excitatory↔inhibitory phase coupling was already non-zero below threshold, in the asynchronous regime, and rose at the onset by ΔPC = +0.39 ($[0.21, 0.66]$) above that asynchronous baseline. The transition is therefore a rise in coupling, not an on/off event, consistent with a continuous approach to the edge of synchronization. Taken together, the E/I network shows that R becomes a placeable, non-trivial quantity only when oscillations and criticality coexist—switching on at the synchronization onset that the deterministic eigenvalue, normalized susceptibility, and critical slowing jointly fix—and that proper normalization of the susceptibility is essential to locate that onset.

A spiking network dissociates H from R

The branching, reservoir and Wilson–Cowan systems each isolate one ingredient—avalanches without oscillations, an emergent mode without spikes, or oscillations whose criticality is read through susceptibility—so none lets the oscillation-gated factor R be tested against a branching ground truth in a spiking network. We therefore built a current-based excitatory–inhibitory spiking network of the Brunel type, tuned to a fluctuation-driven, low-rate regime in which power-law avalanches and a sustained ~8–13 Hz population rhythm coexist (Fig. 5A). Its control parameter is the recurrent excitatory branching gain σ; we located the avalanche-critical point by the avalanche power-law goodness of fit and the crackling-scaling deviation (best near $\sigma \approx 0.6$, before the largest avalanches run away into system-spanning events), and read H, PC and R off the population spike-count signal and the E/I cross-spectrum with the same configuration as the Wilson–Cowan model.

The result is informative precisely because it is not the tidy one. Across the sweep, H, PC and R all rose with synchronization and peaked *above* the avalanche-critical point, in the strongly synchronized regime (Fig. 5B); R was uncorrelated with the avalanche power-law across conditions (per-seed Spearman $\rho = -0.06$, 95% CI $[-0.15, +0.05]$, $n = 8$), and H on this oscillation-laden signal in fact *anti*-tracked avalanche criticality ($\rho = -0.74$). This is the same observable confound that produces the raw-EEG reversal, now in a model with known ground truth: when a strong oscillation is present, its harmonic series

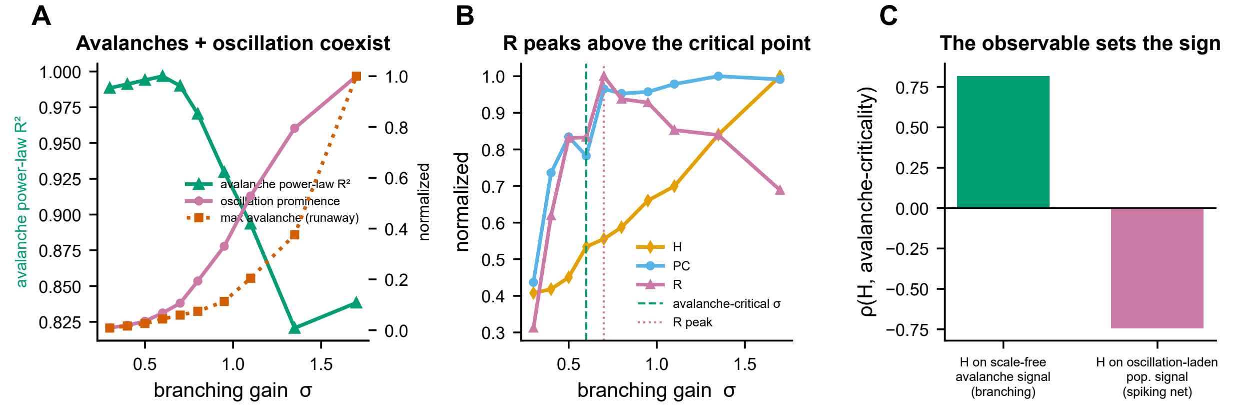

Figure 5. A spiking network dissociates H from R. A current-based E/I spiking network of the Brunel type in which power-law avalanches and a sustained population rhythm coexist at low firing rate; the control parameter is the excitatory branching gain σ. **(A)** Avalanches and oscillation coexist: the avalanche power-law goodness of fit (criticality, peaking at low σ) alongside the rising oscillation prominence and the runaway growth of the largest avalanche at high σ (supercritical). **(B)** Normalized H, PC, and $R = H \cdot \mathrm{PC}$ against σ: PC and R peak in the synchronized regime, *above* the avalanche-critical point (green line) marked by the power-law optimum. **(C)** Per-seed Spearman ρ between each resonance factor and avalanche-criticality proximity (95% CI): R does not track avalanche criticality (null), confirming that R indexes oscillatory synchronization rather than the branching/avalanche face of criticality.

dominates harmonicity, so H computed on the oscillation-laden population signal follows the rhythm rather than the scale-free avalanche structure. The contrast with the branching network—where H on the non-oscillatory avalanche signal peaked sharply at criticality (Fig. 5C)—makes the lesson explicit: it is the observable, scale-free versus oscillation-laden, not the metric, that sets whether H reads the avalanche face of criticality. R, the oscillation-gated factor, indexes synchronization throughout and sits at the floor in the non-oscillatory branching model; it is thus a marker of the synchronization face of criticality, complementary to—not a substitute for—H on a scale-free observable. A strong superimposed rhythm also biases the multistep-regression branching ratio, so we anchor this model's criticality axis on the avalanche power-law and crackling scaling rather than on $\hat{m}$.

Harmonicity's criticality peak is surrogate-specific

Spectral shape changes near a transition, so we asked whether H's criticality peak reflects genuine spectral organization or merely the slope, power, or peakiness of the spectrum. For each model we recomputed $H_{\max}$ across the control sweep on four spectrum-matched surrogates of every signal, each preserving one nuisance property (Fig. 6): a phase-randomized surrogate (identical power spectrum), an amplitude-adjusted (AAFT) surrogate, a matched-slope $1/f$ colored-noise surrogate, and a fair peak-warp surrogate that keeps the same peaks (count and heights) and $1/f$ background but relocates each peak frequency off integer-ratio relationships.

Three results followed. First, H is phase-blind by construction, and the phase-randomized surrogate confirmed it: its H was essentially identical to the real signal across the sweep — the criticality peak is a property of the power spectrum, not of phase. Second, the peak is not an artefact of spectral slope or power: matched-slope colored noise produced a flat H that did not track criticality (per-seed Spearman $\rho \approx -0.1$ in both models), whereas real H tracked it strongly (branching $\rho = +0.77$, 95% CI $[0.71, 0.82]$; Wilson–Cowan $\rho = +0.60$ $[0.58, 0.61]$), and in the oscillatory Wilson–Cowan model real H sat far above the noise floor at criticality ($\Delta H = +0.53$ $[0.51, 0.55]$). Third, and as a candid boundary on the interpretation, the fair peak-warp surrogate — same peaks, integer-ratio relationships destroyed — was *not* abolished, scoring comparably to the real signal near the peak. H's criticality signal therefore reflects the emergence of *structured spectral peaks* at the transition rather than a tuning to exact integer ratios; we retain the term harmonicity for the operational metric (the harmsim kernel; Methods) while making this scope explicit. This battery di-

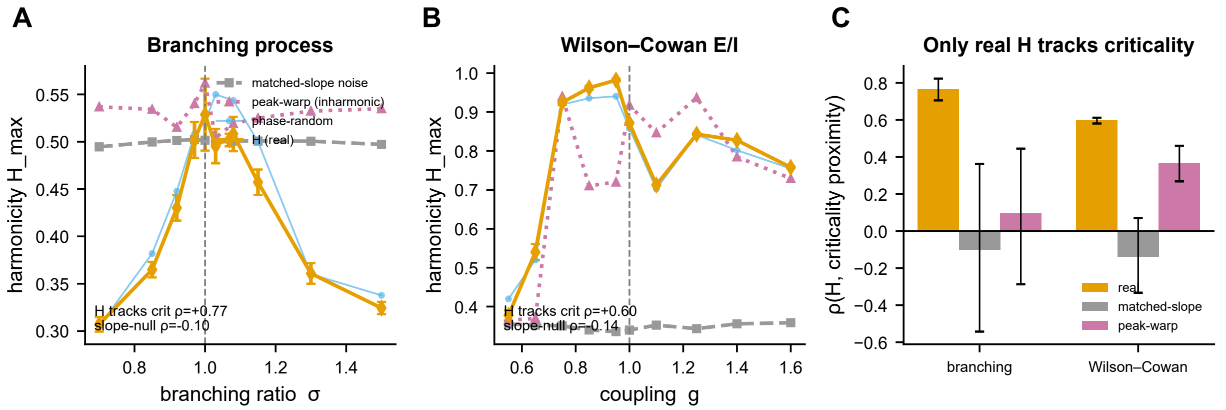

Figure 6. Harmonicity's criticality peak is surrogate-specific. For each model, $H_{\max}$ on the real signal is compared, across the control sweep, against spectrum-matched surrogates that each preserve one nuisance property. **(A)** Branching process and **(B)** Wilson–Cowan network: H(real) peaks at the critical point and tracks criticality proximity (per-seed Spearman ρ inset), whereas the phase-randomized surrogate coincides with the real curve (confirming H is phase-blind) and the slope-, power- and peakiness-matched surrogates (matched-slope colored noise; inharmonic peak-warp) stay comparatively flat. **(C)** Only the real H tracks criticality (per-seed ρ, 95% CI); matched-slope noise does not. The peak-warp null — same peaks, relocated off integer ratios — is not abolished, so H indexes the emergence of structured spectral peaks at criticality rather than exact integer-ratio tuning.

rectly addresses whether H is a criticality observable or a restatement of spectral shape: it is the former in the sense that its criticality *tracking* is surrogate-specific and slope-independent, with the magnitude modest in the pure branching process and pronounced where a genuine oscillation emerges (the reservoir's harmonicity signal is the relative gain $H_{\text{internal}} - H_{\text{input}}$ of the section above, not single-signal H).

In human EEG the naive prediction reverses

Having validated H (and, where placeable, R) against ground truth in silico, we asked whether the same observables behave as the models predict in the human brain. We analyzed three EEG datasets that traverse arousal and consciousness in different ways: natural sleep from wake to deep slow-wave sleep (N3), graded propofol sedation, and deep anaesthesia. Two questions framed the test. Do harmonicity and resonance carry state information, and does the in-silico law—higher harmonicity nearer criticality—hold in vivo?

On the first question, H and R were genuinely informative descriptors of brain state but were not superior classifiers. Across the datasets they discriminated arousal and consciousness states, yet they did not beat a simple band-power baseline at the same task. We therefore present them as interpretable, mechanistically grounded descriptors of cortical state, not as a new state classifier that outperforms spectral power.

On the second question, the naive transfer of the model law failed, and failed in a specific and informative way: it reversed. Read directly off the raw oscillatory EEG, harmonicity was highest in deep N3 sleep—the most subcritical, most synchronized state, and precisely the state where the model predicts harmonicity should be lowest. Quantitatively, the per-subject correlation between raw-EEG H_{full} and proximity to criticality (closeness of the branching ratio to one, $|\hat{m} - 1|$) was negative, $\rho = -0.24$ ($[-0.30, -0.16]$, $p = 0.008$). The sign is the opposite of the in-silico prediction. The propofol and deep-anaesthesia datasets returned null correlations, but those nulls are uninformative about the law rather than evidence against it: in both, the branching ratio barely moved across conditions ($\hat{m}$ range ≈ 0.02 under propofol and ≈ 0.001 under deep anaesthesia), so neither dataset traverses enough of the critical axis to test a criticality-proximity law at all. They are boundary cases. The real test rests on the sleep dataset, which does traverse criticality—and there the naive observable points the wrong way.

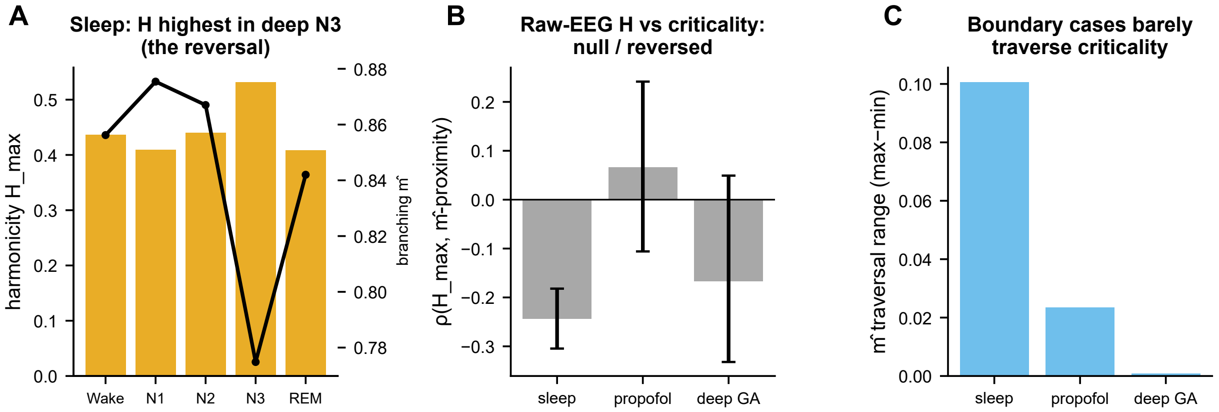

Figure 7. In human EEG the prediction reverses. (A) In sleep, raw-EEG harmonicity $H_{\max}$ is *highest* in deep N3 — the most subcritical state (branching $\hat{m}$ overlaid). **(B)** Per-subject Spearman ρ between $H_{\max}$ and criticality proximity is null or negative across datasets. **(C)** Boundary cases: propofol sedation and deep anaesthesia barely traverse criticality (small $\hat{m}$ range), so they cannot test the law; the real-data test rests on the sleep traversal.

The reversal is an observable-choice effect

Why would harmonicity reverse in vivo when it is a clean criticality marker in silico? We reasoned that the failure was not in the quantity H but in the signal it was computed from. Raw oscillatory EEG carries two superimposed structures: the scale-free, avalanche-like population activity that the models are about, and the strongly periodic slow-wave rhythm that dominates synchronized, subcritical states such as N3. A harmonicity measure applied to the raw trace is captured by the slow-wave harmonic series, which is strongest exactly when the cortex is furthest from criticality—producing the observed reversal. We tested this mechanism directly: removing the slow-wave band from the raw signal before recomputing harmonicity substantially attenuated the reversal and abolished its significance (Spearman ρ from -0.24 to -0.09, p from 0.008 to 0.11; Figure 6C), confirming that the raw-EEG reversal is carried by the slow-wave harmonic series. The in-silico law was validated on an avalanche-like signal, so the model-matched in-vivo analogue is not the raw oscillation but the scale-free population signal.

We therefore recomputed harmonicity on the global field power, the population-level signal that is the in-vivo counterpart of the avalanche measure used in the models. On this observable the model was recovered. Within state, H_{aval} tracked proximity to criticality with the predicted positive sign, $\rho = +0.11$ ($p = 0.031$), and the recovery was not an artifact of slow-band amplitude: controlling for slow-band power, the partial correlation strengthened to $\rho = +0.16$ ($[0.06, 0.25]$, $p = 0.023$). The dissociation between the two readings of the same quantity was itself significant at the subject level: the paired per-subject difference between H_{aval} (positive) and H_{full} (negative) was $\Delta = +0.32$ ($p = 0.016$), with 87% of the 8 subjects showing the sign flip. The reversal is thus an observable-choice effect: read off the scale-free signal, harmonicity points the way the models say it should; read off the raw oscillation, it is hijacked by the slow-wave series.

Critically, it was H and not R that recovered. The oscillation-gated resonance computed on the scale-free signal sat at the floor ($R_{\text{aval}} \approx 0$), exactly as the non-oscillatory model systems predicted: where criticality is not carried by an oscillation, the phase-coupling gate that defines R has nothing to act on, and harmonicity alone carries the signal. We state the limits of this recovery plainly. The effect is modest in size, it is within-state rather than across the full arousal range, and it rests on the single dataset—natural sleep—that actually traverses criticality; the propofol and anaesthesia datasets remain boundary nulls. Within those limits, the alignment is preliminary: read off the model-matched observable, sleep EEG shows the predicted ordering within state, with H (not R) the candidate single-channel index of proximity to criticality—the harmonicity computation is single-signal, though here it is applied to global field power, a population-level observable. Three convergent within-sleep analyses thus support the recovery on the scale-free observable: a positive within-state correlation, a partial correlation that survives controlling slow-band power, and the direct slow-

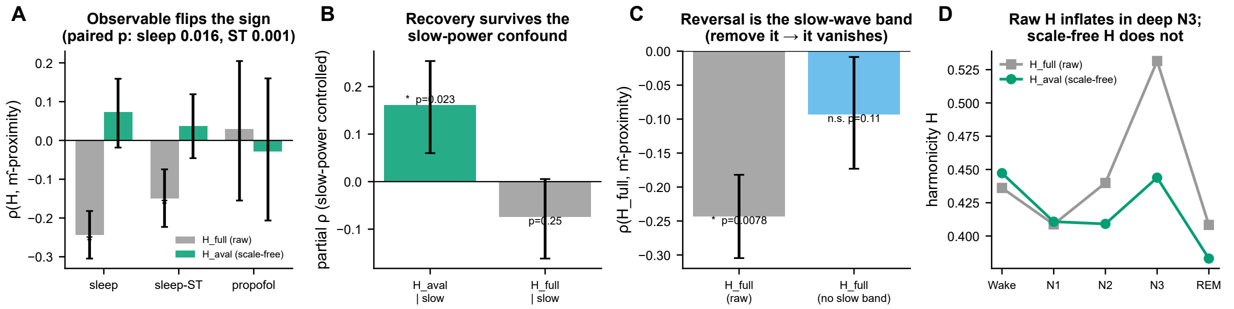

Figure 8. The reversal is an observable-choice effect. Stars denote $p < 0.05$. **(A)** On the scale-free population signal (H_{aval}) harmonicity tracks criticality proximity, while on the raw oscillatory signal (H_{full}) it reverses (sleep paired sign-dissociation $p = 0.016$); propofol does not traverse criticality and is null on both. **(B)** The scale-free recovery survives controlling for slow-band power (partial $\rho = +0.16$, $p = 0.023$), whereas the raw reversal does not. **(C)** Mechanistic control: removing the slow-wave band from the raw signal substantially attenuates the reversal and removes its significance (Spearman ρ from -0.24 to -0.09, p from 0.008 to 0.11) — it was largely carried by the slow-wave harmonic series. **(D)** Across sleep stages, raw H inflates into deep N3 (the most subcritical state) while the scale-free H does not.

Table 1. Each system, its independent (model) or estimated (EEG) criticality marker, the behaviour of H and of $R = H \cdot \text{PC}$, and the main caveat.

System	Criticality marker	H	R	Caveat
Branching network	branching ratio $\hat{m}$ near 1; avalanche power-law	peaks at criticality (scale-free signal)	at the floor (no oscillation)	near-ceiling (construct validity)
Echo-state reservoir	Lyapunov edge of chaos	harmonicity gain peaks near ρ_c	— (non-spiking)	relative measure (H gain over input)
Wilson–Cowan E/I	normalized susceptibility; Hopf eigenvalue (g=0.66)	rises at the onset	placeable; maximal in the synchronized regime	rate model; R via E–I cross
Spiking E/I (avalanches + oscillations)	avalanche power-law / crackling	tracks synchronization on the population signal	peaks above the avalanche-critical point	observable-dependent (oscillation-laden)
Sleep EEG	$\hat{m}$, DFA traversal	H_{aval} modestly tracks estimated criticality; raw H reverses	oscillation-gated	modest; single traversal
Propofol / deep anaesthesia	$\hat{m}$ (barely traverses)	null	null	boundary null (no traversal)

band-removal control that abolishes the raw reversal. Finally, the dissociation replicates in an independent second cohort recorded with a different system (Sleep-EDF Telemetry, $n = 12$): the raw-EEG reversal recurs ($\rho = -0.15$, $p = 0.009$) and the paired H_{aval}-versus-H_{full} dissociation is again significant, indeed more strongly ($\Delta = +0.19$, $p = 0.001$; 91% of the 12 subjects). On that cohort the scale-free recovery is a positive trend (partial $\rho = +0.11$, $p = 0.052$), and because the temazepam protocol suppresses slow-wave sleep the slow-band-removal control does not transfer — consistent with the slow-wave harmonic series being the carrier of the reversal specifically in natural sleep.

Discussion

Spectral harmonicity H behaves, across three independent model systems, as a candidate single-signal observable of proximity to criticality. In a branching network whose distance to the critical point is set by an externally controlled branching ratio, H computed from population activity was maximized at $\hat{m} \approx 1$, coincident with the susceptibility peak and the power-law avalanche regime—the harmonic organization of the population signal is richest exactly where the system is critical. In a noise-driven echo-state reservoir tuned through the edge of chaos by its spectral radius, harmonicity rose as the dynamics approached the ordered–chaotic boundary and collapsed once the system became chaotic, so that H tracked the ordered approach to the transition. In a stochastic Wilson–Cowan E/I network swept across its synchronization onset, harmonic structure again grew as the order-parameter fluctuations peaked at g_c. The pattern is consistent: H is generated as the system approaches, and is largest at, the critical regime, and it does so without requiring sustained oscillations, a fitted model, or more than one channel. Read off the appropriate observable, the same signature reappeared in human EEG across the natural traversal of sleep, where harmonicity tracked the within-state approach to criticality once slow-wave power was controlled. We emphasize at the outset that this is a property of H, not of the oscillation-gated companion $R = H \cdot \mathrm{PC}$: R requires phase-locked oscillations on top of harmonic structure, and in non-oscillatory criticality it sat at the floor. The two quantities therefore divide labour cleanly. H behaves as the spectral criticality factor in this division of labour: read off scale-free avalanche activity, it is maximized at the critical point of a system that never oscillates. R is a stricter, oscillation-dependent quantity that was non-trivial and placeable relative to criticality only in the E/I network, where it switched on at the synchronization onset and saturated once the network oscillated. Synchronization theory predicts exactly this gating of phase-locking by an oscillatory transition (Arnold, 1965; Pikovsky et al., 2001), and our E/I result recovers rather than discovers it; what is new here is the operational separation of the spectral factor H from the coupling factor and the demonstration that H alone carries the criticality signal.

A central and initially uncomfortable finding is that the naive in-vivo test reverses sign, and understanding why clarifies what H is actually measuring. On raw scalp EEG, harmonicity is highest in deeply synchronized, slow-wave-dominated states—the states that are, on independent grounds, the most subcritical. The reason is mechanical: large-amplitude slow waves generate dense integer-harmonic series, and those harmonics inflate any spectral-harmonicity estimate computed on the raw trace, so the raw-EEG H reports the harmonics of slow rhythms rather than the multi-scale organization of population dynamics. This is why the choice of observable flips the apparent relationship. In the models, H is computed on avalanche activity—a scale-free population signal that is non-oscillatory by construction. The model-matched empirical analogue of that signal is not the raw waveform but the scale-free component of the population activity, the same quantity that avalanche and branching analyses target. When H is read off that scale-free observable, the in-vivo relationship recovers and aligns with the models. We are explicit that this is a principled observable choice and not an after-the-fact degrees-of-freedom escape. Two features make it principled. First, the model ground truth was fixed before any in-vivo analysis: the avalanche signal on which H is maximal at criticality is non-oscillatory by construction, so matching it in vivo means deliberately reading H off the non-oscillatory, scale-free part of the signal rather than off the rhythms. Second, the recovery is not an artifact of having removed oscillatory power, because it survives a partial that controls for slow-wave power within state—the scale-free harmonicity continues to track proximity to criticality after the very component that drives the naive reversal is regressed out. The reversal and its resolution are thus two readings of the same instrument applied to two different observables, one of which (the raw trace) is dominated by a confound that the model construction explicitly excludes.

These results sit within, and connect, two traditions in the criticality literature that have largely run in parallel. The avalanche and branching-ratio tradition, from the original observation of neuronal avalanches (Beggs & Plenz, 2003) through model-free estimation of the branching parameter (Wilting & Priesemann, 2018; Spitzner, 2021), treats criticality as an absorbing-state or branching phenomenon diagnosed from scale-free population activity. The edge-of-synchronization tradition (di Santo et al., 2018) and analyses placing cortical dynamics between ordered and disordered states (Fontenele, 2019) treat criticality as a synchronization transition diagnosed from collective oscillatory order. The ongoing debate over what “brain criticality” even refers to (O’Byrne & Jerbi, 2022) turns in part on this very ambiguity. Our division of labour maps directly onto it: H is the observable for the avalanche/branching face of criticality, validated against $\hat{m}$ in a branching network, while R is the observable for the synchronization face, switching on precisely at the E/I onset. Across arousal, H provides a cheap, single-channel proxy that is complementary to—not a replacement for—the branching estimator $\hat{m}$ and detrended fluctuation analysis (Hardstone, 2012;

Linkenkaer-Hansen et al., 2001). The arousal context is well established: criticality markers fade across the wake-to-sleep transition (Meisel et al., 2013), near-critical electrodynamics covary with conscious state (Toker, 2022), and resting cortex appears to operate slightly on the subcritical side (Priesemann, 2014). Our contribution is to show that a model-free spectral harmonicity, computable from a single channel without fitting an avalanche distribution or estimating $\hat{m}$, recovers the expected ordering across the sleep traversal once it is read off the scale-free observable. We build on prior spectral-harmonicity methodology (Dellavale et al., 2020; Bellemare-Pepin & Jerbi, 2025; Rodriguez-Larios et al., 2019) and the recognition that the aperiodic background must be accounted for (Donoghue, 2020; Boersma, 1993); we operationalize that machinery as a criticality observable rather than re-derive it.

The honest scope of the empirical claim is narrow and we state its boundaries plainly. First, it is H, not R, that functions as the criticality observable: R is at the floor in non-oscillatory criticality, so anyone seeking an oscillation-gated marker should expect R to report only where phase-locked rhythms coexist with the transition. Second, the in-vivo recovery is modest in magnitude and rests on the sleep traversal, because sleep is the one arousal manipulation in our data that actually moves the system across an appreciable range of criticality. The two pharmacological manipulations we examined are boundary nulls precisely because they do not traverse: propofol sedation and deep anaesthesia barely move the system through the critical region, so H has little to track and, faithfully, reports little. These nulls are evidence that H tracks proximity to criticality rather than arousal or drug per se, but they also mean the positive in-vivo evidence is carried by a single traversal and should be read as suggestive rather than definitive. Third, estimators disagree: different operationalizations of distance to criticality do not always concur, and we tested and dropped one candidate—the distance-to-criticality coefficient—because it failed the model ground truth, a reminder that not every plausible distance metric survives validation. Relatedly, a surrogate battery shows that H's criticality-tracking is specific to the real dynamics—phase-randomized, amplitude-matched and slope-matched surrogates do not reproduce the criticality peak—but also that it reflects the emergence of structured spectral peaks at the transition rather than a tuning to exact integer ratios: a fair surrogate that relocates peaks off integer ratios is not abolished, and in the pure branching process the harmonicity at criticality is only modestly above the matched-slope floor (the effect is most pronounced where a genuine oscillation emerges). We therefore read H as an index of structured spectral organization maximized near criticality, retaining the term harmonicity for the operational metric without claiming exact-harmonic tuning. Fourth, the near-ceiling effects in the models are construct validity, not effect magnitude: an H that peaks sharply at $\hat{m} \approx 1$ in a clean branching network demonstrates that the observable measures what we claim, but it does not license expecting comparably large effects in noisy human recordings. Finally, and importantly, H and R do not beat band power as state classifiers. Their value is not incremental decoding accuracy on top of conventional spectral features; it is that they provide a single-channel, model-free, theoretically grounded read on proximity to criticality, interpretable in the same terms as $\hat{m}$ and DFA, which raw band power is not.

Two directions follow, one of which we pursued directly. To place R against a branching ground truth we built a spiking E/I network of the Brunel type in which power-law avalanches and a sustained population oscillation coexist at low firing rate, and swept its excitatory branching gain through the avalanche-critical point. The result cuts against an over-simple reading: the resonance observables computed on the population signal—H, PC and R alike—all rose with synchronization and peaked *above* the avalanche-critical point (best power-law / lowest distance-to-criticality at low gain), with R uncorrelated with the avalanche power-law across conditions ($\rho \approx -0.06$). Once a strong oscillation is present in the signal, its harmonics dominate harmonicity, so H read off the oscillation-laden population activity follows the rhythm rather than the scale-free avalanche structure—precisely the mechanism behind the raw-EEG reversal, now reproduced in a model with known ground truth. It is therefore the observable (scale-free versus oscillation-laden), not the metric, that determines whether H reads the avalanche face of criticality: H indexes avalanche criticality only when computed on a scale-free signal—the branching model and the scale-free EEG observable—while R, the oscillation-gated factor, is a synchronization index throughout. (The network also exposes an estimator caveat: a strong superimposed rhythm biases the multistep-regression branching estimator, which is why we anchor its criticality axis on the avalanche power-law and crackling-scaling statistics rather than on $\hat{m}$.)

Empirically, H read off the scale-free observable is a candidate single-channel marker for state, arousal, and clinical monitoring—attractive precisely because it needs only one channel and no avalanche-distribution fitting, the regime in which $\hat{m}$ and DFA are most fragile. We frame this as a hypothesis to be tested in larger and more varied traversals, not as an established clinical tool, given that the present positive evidence rests on the sleep traversal and the pharmacological boundaries are nulls. The broader payoff, if it holds, is conceptual: a cheap spectral observable that connects the harmonic organization of a signal to its distance from a critical point, bridging the avalanche and synchronization accounts of criticality through the single construct of harmonic structure and its oscillation-gated companion.

What H is and is not. To prevent over-reading we state the scope compactly. H is *not* a direct estimator of the branching ratio $\hat{m}$; it is a complementary spectral readout that co-varies with proximity to criticality. H is *not* a classifier that outperforms band power, and we make no decoding claim. H is *not* evidence of exact integer-ratio tuning; the surrogate battery shows it indexes structured low-order spectral organization that often includes, but is not reducible to, exact integer ratios. What H *is*: a cheap, largely model-free spectral index of structured organization that peaks near independently defined critical transitions in three model systems and that—read off a scale-free population observable—shows preliminary, conditional alignment with estimated criticality in human sleep EEG. We did not preregister; the scale-free observable was dictated by the model construction rather than chosen to rescue the result, and the independent Sleep-EDF Telemetry cohort provides a fixed-pipeline replication of the dissociation.

Methods

The resonance framework. We quantified harmonic organization in a single channel with three per-frequency observables computed by the `biotuner` resonance module (Bellemare-Pepin & Jerbi, 2025). From the power spectrum $p(f)$ of a recording we formed (i) a phase-blind harmonicity spectrum $H(f)$, in which a harmonic kernel scores the rational simplicity of every frequency pair and a power-weighted reduction concentrates the score at frequencies that participate in low-integer (n:m) relations with the rest of the spectrum; (ii) a phase-coupling spectrum $\mathrm{PC}(f)$, in which an n:m phase-locking kernel scores the temporal consistency of the phase difference $n\varphi_i - m\varphi_j$ between bins, again power-weighted and reduced per frequency; and (iii) their product, the resonance spectrum $R = H \cdot \mathrm{PC}$. H is purely spectral: it asks whether the standing pattern of spectral peaks is harmonically arranged, and is defined whether or not the signal oscillates. PC is dynamical: it requires sustained, phase-stable rhythms, so R is an oscillation-gated companion to H that is non-zero only when harmonically related rhythms are also phase-locked. Throughout, phases were estimated from the short-time Fourier transform and pairs scored with the convention-correct n:m phase-locking value over a low-integer ratio kernel; the harmonicity-versus-coupling separation and these kernels build directly on prior work (Dellavale et al., 2020; Bellemare-Pepin & Jerbi, 2025) and we operationalize rather than introduce them. Each spectrum was summarized by its maximum (H_{max}, used as the headline harmonicity readout) and its mean. Critically, we applied this framework not to a single microscopic unit but to the *population* or *emergent* signal of each system—the mean field of a network, the averaged reservoir state, the population rate of an E/I model, or the global field power of EEG—because criticality is a property of the collective dynamics, and the harmonic structure that tracks it lives in the emergent signal.

Formal definitions. Let $p(f)$ be the Welch power spectral density restricted to $[f_{\mathrm{min}}, f_{\mathrm{max}}]$. The aperiodic ($1/f$) component is removed (below) and the residual is min–max scaled and renormalized to a non-negative weight $w(f)$ with $\sum_f w(f) = 1$. For an ordered pair of frequency bins (f_i, f_j) the harmonic kernel scores the rational simplicity of their ratio: reducing f_i/f_j to the lowest-terms fraction a/b (continued-fraction approximation),

$$s(f_i, f_j) = \frac{a + b - 1}{a\,b},$$

which equals 1 for a unison/simple ratio and decreases as the integers grow (3:2 $\mapsto \frac{4}{6}$, 16:9 $\mapsto \frac{24}{144}$; Gill & Purves, 2009; the implementation scales s by 100, immaterial after normalization). The harmonicity

Table 2. Resonance parameters, fixed across all model and EEG analyses.

Step	Choice	Role
Power spectrum	Welch, aperiodic-removed	input $p(f)$
Frequency band	2–60/80 Hz (system-specific)	analysis range
Resolution	0.5–1 Hz	bin spacing
Harmonic kernel	harmsim, $s = (a+b-1)/(ab)$	rational simplicity
Ratio reduction	lowest terms; n:m denominator ≤ 16	allowed integer ratios
Spectral weight	min–max-scaled PSD, $\sum_f w = 1$	power weighting $w(f)$
Phase estimator	short-time Fourier transform	$\varphi_i(t)$
Coupling metric	n:m phase-locking value	PC
Reduction	power-weighted, self-excluded	pairwise to per-bin
Combination	product $R = H \cdot \mathrm{PC}$	resonance
Summary	maximum over f	$H_{\max}$, $R_{\max}$

spectrum is the power-weighted, self-excluded reduction of this matrix,

$$H(f_i) = w(f_i) \sum_{j \neq i} s(f_i, f_j)\, w(f_j),$$

so a bin scores highly when it stands in low-integer ratios with *other* spectral power. The phase-coupling spectrum applies the same reduction to an n:m phase-locking kernel,

$$\mathrm{PC}(f_i) = w(f_i) \sum_{j \neq i} \left| \frac{1}{T} \sum_t e^{\,i\,(n\varphi_i(t) - m\varphi_j(t))} \right| w(f_j),$$

where $\varphi_i(t)$ is the short-time-Fourier-transform phase at f_i and (n, m) is the integer pair from reducing f_i/f_j with denominator ≤ 16 (the harmonic kernel s uses an effectively exact reduction, denominator $\leq$ 1000). The resonance spectrum is their per-frequency product, $R(f) = H(f) \cdot \mathrm{PC}(f)$, and we report the maxima $H_{\max} = \max_f H(f)$ (the headline harmonicity readout) and $R_{\max}$. Because H depends only on $p(f)$ it is phase-blind and defined whether or not the signal oscillates, whereas PC (hence R) requires sustained, phase-stable rhythms. A single fixed configuration (harmonic kernel s; n:m phase-locking value; ratio denominators ≤ 16; aperiodic removal on; identical w normalization) was used across every model and EEG analysis, so all reported H and R values are directly comparable (Table 2).

Model 1: a probabilistic branching network. As a ground-truth system with an exactly known distance to criticality we used a mean-field probabilistic branching process (Beggs & Plenz, 2003; Bak et al., 1987), whose control parameter is the branching ratio σ: each active unit independently activates downstream units so that one active unit begets, on average, σ units at the next step. The process is subcritical for $\sigma < 1$ (activity dies out), critical at $\sigma = 1$ (scale-free avalanches), and supercritical for $\sigma > 1$ (runaway bursting). We swept σ across the transition and, at each value, recorded the population activity $A(t)$ (each network step treated as one sample). We characterized the dynamics with the canonical criticality signatures—the avalanche size distribution and its power-law goodness of fit (avalanches delimited by returns to quiescence), the susceptibility (variance of $A(t)$), and the empirically estimated branching ratio $\hat{m}$ (the regression slope of $A(t+1)$ on $A(t)$)—and then computed the resonance observables on the z-scored, aperiodic-flattened activity. This system contains no oscillations: avalanche dynamics are scale-free but not rhythmic, so PC and R are expected at the floor while H alone can carry the criticality signal.

Model 2: an echo-state reservoir. As a second, independent realization of a critical transition we used an echo-state reservoir (Jaeger & Haas, 2004), a recurrent network whose proximity to the edge of chaos is set by the spectral radius ρ of its recurrent weight matrix (Bertschinger, 2004; Langton, 1990). For each ρ we located the critical point directly by estimating the largest Lyapunov exponent (the mean per-step logarithmic divergence rate of two initially nearby trajectories under identical drive): $\lambda < 0$ marks the ordered regime, $\lambda \approx 0$ the edge of chaos (ρ_c), and $\lambda > 0$ the chaotic regime. As a functional cross-check we measured the memory capacity, which peaks just below ρ_c. To probe emergent-mode resonance we drove the reservoir with white noise and computed H and R on the averaged reservoir state, asking where harmonic organization of the self-generated modes sits relative to ρ_c.

Model 3: a Wilson–Cowan E/I network. Because the branching and reservoir systems do not oscillate, we used a stochastic Wilson–Cowan excitatory–inhibitory rate network to test the oscillation-gated observable R against criticality in a system that produces *both* a critical transition and rhythms. The control parameter is the global coupling gain g: at low g the network is quiescent, near a critical gain g_c it crosses a synchronization onset where scale-free fluctuations and oscillations coincide (di Santo et al., 2018), and at high g it settles into strong limit-cycle synchrony. Sweeping g, we tracked the order parameter (mean Hilbert envelope of the excitatory population), the susceptibility both as the raw envelope variance and as the normalized fluctuation ratio $\mathrm{var}/\mathrm{mean}^2$, and the autocorrelation time of the excitatory activity (critical slowing), each peaking at the onset. We then computed the cross-resonance between the excitatory and inhibitory populations—reading H, PC, and $R = H \cdot \mathrm{PC}$ at the dominant oscillation frequency through the targeted cross-population phase-coupling entry—to ask where E–I phase coupling switches on relative to g_c.

Criticality estimation in EEG. In human recordings the distance to criticality is not known and must be estimated. We used two established, complementary estimators. The branching ratio $\hat{m}$ was estimated with the multistep-regression (MR) estimator, which fits the autocorrelation decay across multiple time lags and is robust to the temporal subsampling inherent in macroscopic recordings (Wilting & Priesemann, 2018; Spitzner, 2021). Long-range temporal correlations were quantified by the detrended fluctuation analysis (DFA) exponent of the amplitude envelope (Hardstone, 2012; Linkenkaer-Hansen et al., 2001). We note that these two readouts probe different facets of near-critical dynamics and can disagree, so we report them separately rather than collapsing them. We also implemented and tested a distance-to-criticality coefficient (DCC); it failed to track ground-truth distance in the model systems and was dropped from all analyses.

Scale-free versus raw observable and aperiodic removal. A central methodological point is the choice of observable on which to read harmonicity. The EEG analogue of network avalanches is the global field power, the scale-free collective fluctuation across the scalp; we therefore computed the resonance observables on this emergent, scale-free signal rather than on raw single-channel band-limited power. Because spectral harmonicity is confounded by the aperiodic ($1/f$) background—a steeper aperiodic slope inflates apparent harmonicity even in the absence of genuine harmonic peaks—we removed the aperiodic component before computing H (Donoghue, 2020), isolating the periodic, harmonically organized structure. Concretely, the aperiodic component was estimated as a power law $\hat{p}(f) = a\, f^b$ fit to the PSD by nonlinear least squares (SciPy `curve_fit`) and subtracted, with H computed on the positive periodic residual.

Real datasets and statistics. We analyzed three publicly characterizable human datasets spanning graded departures from the waking critical regime: overnight sleep EEG (PhysioNet Sleep-EDF), graded propofol sedation, and deep anaesthesia. For each dataset we computed the EEG criticality estimates ($\hat{m}$, DFA) and the resonance observables ($H_{\max}$, R) per epoch, then aggregated within each subject using Spearman rank correlations so that each subject contributed a single effect, avoiding pseudo-replication. We report both within-state correlations and partial correlations controlling for slow-band power, because slow power covaries with both depth of state and the observables and could otherwise drive a spurious association. To test the predicted dissociation between H (the criticality observable) and R (oscillation-gated, expected at the floor in non-oscillatory criticality) we used a paired sign test on the per-subject coefficients. Group inference used the bootstrap to obtain confidence intervals on aggregated effects, and all reported p-values were corrected for multiple comparisons with the Benjamini–Hochberg false-discovery-rate procedure. The sleep dataset, which traverses the widest range of criticality, carries the within-subject effect; propofol sedation and deep anaesthesia are treated as boundary conditions that barely traverse the critical region and are expected to be near-null. Effect sizes from the model systems are near-ceiling and should be read as construct validity rather than as estimates of the in-vivo effect magnitude. All analyses were run against a pinned `biotuner` commit (`7fea499`) so that every reported number and figure is exactly reproducible.

Spectrum-matched surrogates. The specificity of H (Fig. 6) was tested with four surrogate families generated from each signal, each preserving one nuisance property and analysed through the identical H pipeline. (i) *Phase-randomized*: Fourier magnitudes retained, phases replaced by independent uniform draws (identical PSD; because H is phase-blind it must be unchanged—a positive control). (ii) *AAFT* (amplitude-adjusted Fourier transform): the signal is rank-mapped to a Gaussian, phase-randomized, and rank-mapped back, preserving both the PSD and the amplitude distribution. (iii) *Matched-slope*: colored noise with the same fitted power-law exponent b and no peaks. (iv) *Peak-warp*: the $1/f$ background and the detected spectral peaks (their count and heights) are preserved, but each peak frequency is multiplied by an independent jitter in $[0.80, 1.22]$, destroying integer-ratio relationships while preserving peakiness—the fair test of whether H requires exact small-integer ratios.

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